

Problem

Indonesia carries **10% of the world’s TB burden** (WHO Global TB Report 2025). About **200,000 people per year transmit TB before ever being detected**, and screening is bottlenecked by chest-X-ray reading capacity at the last mile. Root cause: **the care loop moves slower than transmission**.

What we built — one calibrated risk state, three interfaces

- 1 • **Screen**. A chest X-ray becomes a calibrated risk with a *deferral band*: when model families disagree, SIGAP says “uncertain — human read” instead of forcing a verdict.
- 2 • **Refer**. The score maps to an action band (refer today / re-read / educate) and a printable referral letter for GeneXpert confirmation.
- 3 • **Support**. The same risk state follows the patient home: adherence check-ins update the score and trigger escalation.

Demo: two real external cases

REFER TB+ X-ray → p(TB)=0.98, disagreement u=0.01

UNCERTAIN normal X-ray from an unseen site → p(TB)=0.32, u=0.37 → “human re-read”

English + Indonesian UI; **web-first responsive** (desktop screening workspace, mobile parity) for community health workers; <3 minutes per patient; audit-friendly outputs.

Evidence — we report the number most projects hide

Evaluation	AUROC	sens@spec90	Calibration (ECE)
Internal random split (<i>the illusion</i>)	0.98 – 0.9999	—	—
External hospitals — supervised fine-tune (EfficientNet-BO + ConvNeXt-T)	0.675 [0.636–0.712]	0.27	0.21
External — RAD-DINO self-supervised probe (final model)	0.887 [0.862–0.909]	0.71	0.25

Data audit before training. Perceptual-hash de-duplication removed **338 internal duplicates** and **19 training images that leaked into the external test set**; the frozen split manifest is published.

Representation study (3 families). Domain-pretrained SSL (RAD-DINO, chest X-ray) beats supervised fine-tuning by **+0.21 AUROC** on unseen hospitals, while stacking/ensembling across families *hurts* (tested, reported).

Honesty & limitations

- Research triage prototype — **not a diagnostic device**; GeneXpert confirmation stays mandatory.

WHO target (sens > 90%) is **not yet met**; reported as-is (0.71 at spec 0.90).
- No Indonesian data in training; site thresholds do not fully transfer — measured over-triage is reported, not hidden.

Retrospective public data (radiographic labels), aggressive re-processing may evade de-duplication.

Stack

Model	RAD-DINO (SSL, chest X-ray) + linear probe · temperature scaling · cross-family disagreement deferral
Engineering	PyTorch / timm / scikit-learn · FastAPI serving + Grad-CAM · Next.js (ID-first) · Kaggle T4 training via API
Reproducibility	Public repo: frozen data manifest, audit reports, external evaluation scripts, model configs